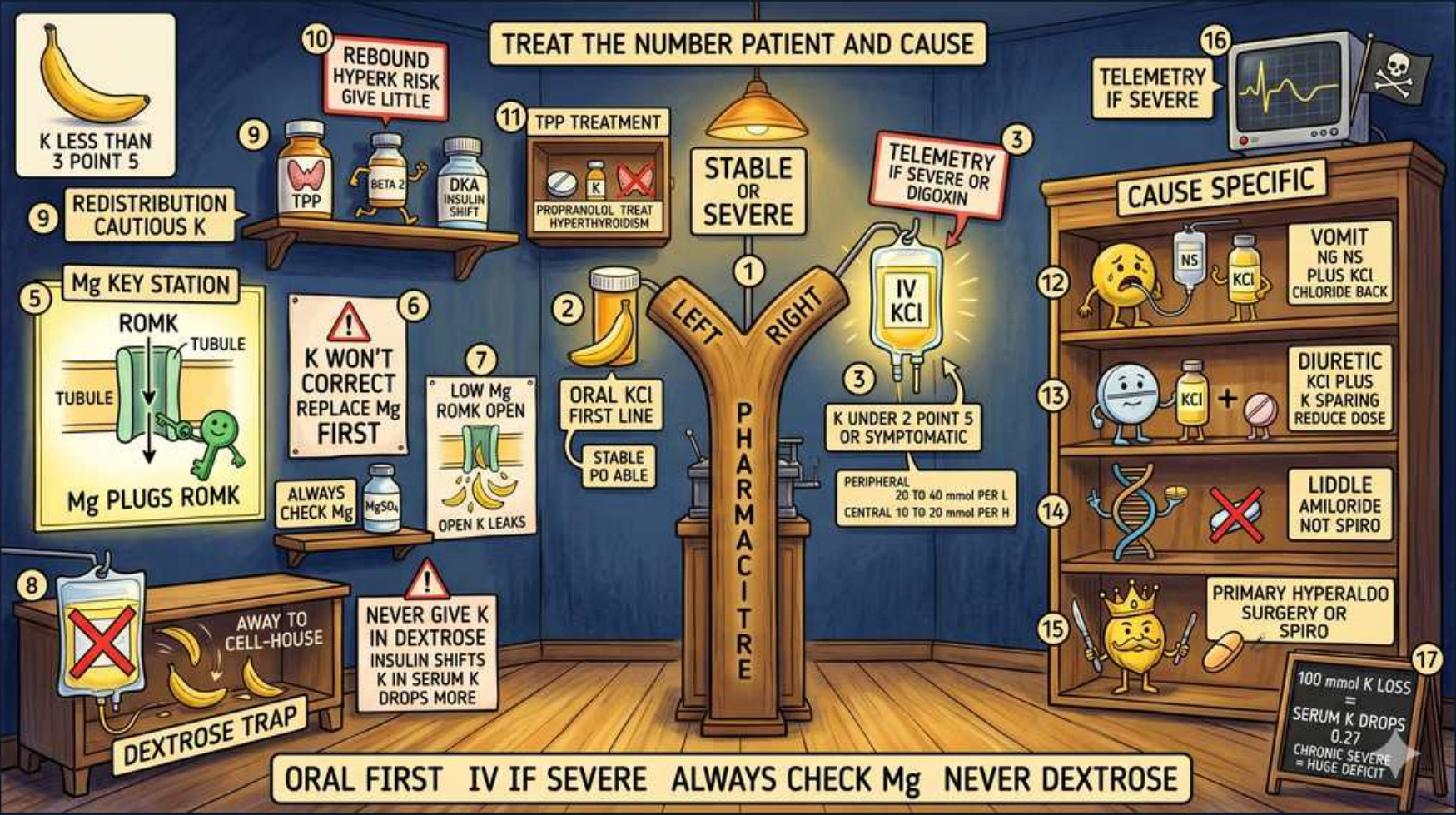


Hypokalemia — treatment



1. Treat patient AND cause

WHAT YOU SEE

Banner: TREAT THE NUMBER, PATIENT AND CAUSE; bottom: oral first, IV if severe, always check Mg, never dextrose

WHAT IT ENCODES

Four pillars of repletion: (1) oral KCl is first line, (2) IV only if severe/symptomatic or NPO, (3) ALWAYS check and replace Mg, (4) never dilute K in dextrose alone. Treat the underlying cause (vomiting, diuretic, hyperaldo) — repletion alone fails if losses continue. As a rule of thumb each 10 mEq of K raises serum K by ~0.1 mEq/L, but the total-body deficit is far larger than the serum number implies.

BOARD TRAP

WRONG: "Just give IV potassium fast to a stable patient with K 3.0." Stable, asymptomatic, able-to-swallow patients get ORAL KCl — IV is reserved for severe (<2.5), symptomatic, arrhythmia, or NPO patients. The discriminator is symptoms/severity and PO access, not the absolute number alone.

2. Stable or severe fork

WHAT YOU SEE

Pharmacure tree splitting LEFT (oral) and RIGHT (IV)

WHAT IT ENCODES

The treatment route forks on severity and PO access. Stable + able to take PO → oral KCl (left branch). Severe (K <2.5), symptomatic (weakness, ileus, arrhythmia, ECG changes), or unable to swallow → IV KCl (right branch). Both arms still require Mg checking and cause-directed therapy.

BOARD TRAP

WRONG: choosing the route purely off the lab value. A patient with K 2.3 who is asymptomatic, eating, and hemodynamically stable can often start oral repletion; conversely a patient at 3.0 on digoxin with PVCs needs IV. Severity + symptoms + PO access determine the branch, not the number in isolation.

3. Oral KCl first line

WHAT YOU SEE

Left branch: ORAL KCl first line; stable PO able

WHAT IT ENCODES

Oral KCl is preferred for mild-to-moderate hypokalemia (K roughly 2.5–3.5, no severe symptoms). It is safe, cannot cause an iatrogenic bolus, and the gut self-limits absorption. Typical dosing is 20–40 mEq per dose, repeated; potassium chloride is the salt of choice in most cases (corrects the common coexisting metabolic alkalosis/chloride loss).

BOARD TRAP

WRONG: "Use potassium bicarbonate/citrate for the vomiting patient with metabolic alkalosis." In chloride-depletion alkalosis (vomiting, NG suction, diuretics) you want KCl, which replaces both K and Cl. Reserve K-citrate/bicarbonate for hypokalemia WITH a metabolic acidosis (e.g., RTA, diarrhea).

4. IV KCl + rates

WHAT YOU SEE

Right branch: IV KCl; K under 2.5 or symptomatic; peripheral 10, central 10–20 mmol per hr

WHAT IT ENCODES

IV KCl rate limits: peripheral line up to ~10 mEq/h (higher concentrations cause phlebitis/pain); central line up to ~10–20 mEq/h with continuous cardiac monitoring. Maximum concentration is roughly 10 mEq/100 mL peripherally, up to 20–40 mEq/100 mL centrally. Use saline as the diluent, never dextrose.

BOARD TRAP

WRONG: running 40 mEq/h peripherally to "fix it fast," or giving an IV K "bolus." Exceeding safe rates risks fatal hyperkalemia and cardiac arrest; high peripheral concentration causes phlebitis. There is no such thing as an IV potassium push — it is always a controlled infusion with monitoring.

5. Telemetry if severe

WHAT YOU SEE

Telemetry monitor, skull: telemetry if severe

WHAT IT ENCODES

Severe hypokalemia, symptomatic patients, ongoing IV repletion, or any patient on digoxin warrants continuous ECG/telemetry. Hypokalemia prolongs repolarization: ECG shows flattened T waves, U waves, ST depression, and QT/QU prolongation, predisposing to torsades and (with digoxin) to digoxin toxicity arrhythmias.

BOARD TRAP

WRONG: skipping telemetry in a digoxin patient because the K is "only mildly low." Hypokalemia markedly potentiates digoxin binding to the Na/K-ATPase and precipitates life-threatening arrhythmias even at modest K reductions — these patients must be monitored and aggressively repleted.

6. Mg key station

WHAT YOU SEE

Mg KEY STATION: ROMK tubule, Mg plugs ROMK; always check Mg

WHAT IT ENCODES

Intracellular magnesium normally plugs the ROMK channel in the distal nephron, limiting K secretion. When Mg is depleted, ROMK is disinhibited (left open) and the kidney wastes K continuously. This is why Mg status must be checked and corrected in every hypokalemic patient — it is the rate-limiting step in refractory cases.

BOARD TRAP

WRONG: ordering only a potassium level and ignoring magnesium. Common causes (loop/thiazide diuretics, diarrhea, alcoholism, PPIs, cisplatin) deplete BOTH ions, and you will chase your tail repleting K while Mg-driven renal K wasting continues.

7. K won't correct — replace Mg first

WHAT YOU SEE

Sign: K won't correct, replace Mg first; low Mg ROMK open / K leaks

WHAT IT ENCODES

Refractory hypokalemia is hypomagnesemia until proven otherwise. With low Mg, ROMK stays open and renal K loss persists despite aggressive repletion. Correct Mg FIRST (e.g., IV magnesium sulfate 1–2 g) and the K will then respond; in coexisting deficiency repletion of both is given in parallel.

BOARD TRAP

WRONG: "K isn't rising despite 200 mEq — give even more potassium." The board answer in a diuretic/alcoholic patient whose K won't budge is to CHECK AND REPLACE MAGNESIUM. Escalating K alone fails and risks rebound hyperkalemia once Mg is finally corrected.

8. Dextrose trap

WHAT YOU SEE

DEXTROSE TRAP: never give K in dextrose; insulin shifts K in, serum K drops more

WHAT IT ENCODES

Never use dextrose-containing fluid as the K diluent: glucose triggers endogenous insulin release, which drives K intracellularly and can transiently DROP serum K further before repletion catches up. Mix KCl in normal saline (or half-normal saline) instead.

BOARD TRAP

WRONG: choosing "KCl in D5W" for an acutely symptomatic hypokalemic patient. The insulin surge from dextrose shifts K into cells and acutely worsens the hypokalemia — the correct diluent is saline. (Note: this is the opposite scenario from treating hyperkalemia, where insulin+dextrose is therapeutic.)

9. Rebound hyperK risk

WHAT YOU SEE

REBOUND HYPERK RISK, give little (TPP, beta, DKA bottles)

WHAT IT ENCODES

In pure transcellular SHIFT states — thyrotoxic/hypokalemic periodic paralysis, beta-agonist excess, DKA before fluids/insulin equilibrate — total body K is normal and the K is merely hiding inside cells. Give SMALL, cautious doses: when the shift reverses, the K floods back out and aggressive repletion causes dangerous rebound hyperkalemia.

BOARD TRAP

WRONG: treating periodic-paralysis hypokalemia like a true deficit with large K loads. Once the trigger resolves the intracellular K re-emerges and the patient overshoots into hyperkalemia — give modest K and treat the underlying shift instead.

10. TPP treatment

WHAT YOU SEE

TPP TREATMENT: propranolol, treat hyperthyroidism

WHAT IT ENCODES

Thyrotoxic periodic paralysis (classically a young Asian male with acute proximal weakness and hyperthyroidism) is a shift phenomenon. Give non-selective beta-blockade — propranolol — to block the beta-adrenergic Na/K-ATPase-driven K shift, give only modest K, and definitively treat the hyperthyroidism. Avoid large K doses (rebound hyperK).

BOARD TRAP

WRONG: "Load the TPP patient with high-dose IV potassium." Because the K is shifted (not lost), high doses cause rebound hyperkalemia when the attack breaks. Propranolol plus low-dose K plus treating the thyroid is the answer.

11. Cause-specific: vomiting

WHAT YOU SEE

Cause shelf: VOMIT — NS plus KCl, replace chloride back

WHAT IT ENCODES

Vomiting/NG suction causes a chloride-responsive metabolic alkalosis with volume depletion and secondary hyperaldosteronism driving renal K loss. Treat with normal saline (restores volume and chloride, switches off aldosterone) PLUS KCl. Urine chloride is low (<20), confirming saline-responsiveness.

BOARD TRAP

WRONG: giving potassium without volume/chloride repletion in the vomiting patient. Until you restore Cl and volume, secondary hyperaldosteronism keeps wasting K — saline + KCl together, not KCl alone.

12. Cause-specific: diuretic

WHAT YOU SEE

DIURETIC — KCl plus K-sparing, reduce dose

WHAT IT ENCODES

Loop and thiazide diuretics cause renal K (and Mg) wasting. Manage by repleting KCl (and Mg), reducing the diuretic dose if possible, and adding a K-sparing agent (amiloride, triamterene, or spironolactone) to offset ongoing loss. Always check Mg, which is commonly co-depleted.

BOARD TRAP

WRONG: endlessly repleting oral K in a patient on a high-dose loop diuretic without addressing the drug. The durable fix is dose reduction and/or a K-sparing diuretic, plus correcting Mg — otherwise the wasting continues.

13. Liddle vs primary aldo tx

WHAT YOU SEE

LIDDLE — amiloride not spiro; PRIMARY HYPERALDO — surgery or spironolactone

WHAT IT ENCODES

Liddle syndrome is a gain-of-function ENaC mutation (constitutively open sodium channel) causing HTN + hypokalemia + metabolic alkalosis with LOW aldosterone and LOW renin — treat with an ENaC blocker (amiloride or triamterene). Primary hyperaldosteronism (Conn) has HIGH aldosterone/low renin — treat with adrenalectomy (adenoma) or a mineralocorticoid-receptor antagonist (spironolactone/eplerenone) for bilateral hyperplasia.

BOARD TRAP

WRONG: giving spironolactone for Liddle syndrome. Liddle is a channel defect with LOW aldosterone, so an aldosterone antagonist won't work — the answer is amiloride/triamterene (block ENaC directly). Spironolactone is correct only when aldosterone is the problem (primary hyperaldo).

14. Severe deficit magnitude

WHAT YOU SEE

Note: 100 mmol K loss $\approx$ serum K drop $\sim$ 0.3

WHAT IT ENCODES

Serum K vastly underestimates total-body deficit because $\sim$ 98% of body K is intracellular. As a guide, every $\sim$ 0.3 mEq/L fall in serum K reflects roughly a 100 mEq total-body deficit, so a K of 3.0 may represent a 200–400 mEq shortfall. Repletion therefore takes more than the bare number suggests — except in pure shift states, where total body K is normal.

BOARD TRAP

WRONG: stopping repletion the instant serum K crosses into the normal range in a truly depleted patient. The intracellular compartment is still empty; under-replacement leads to recurrent hypokalemia. (The exception: shift states, where the deficit is apparent, not real.)